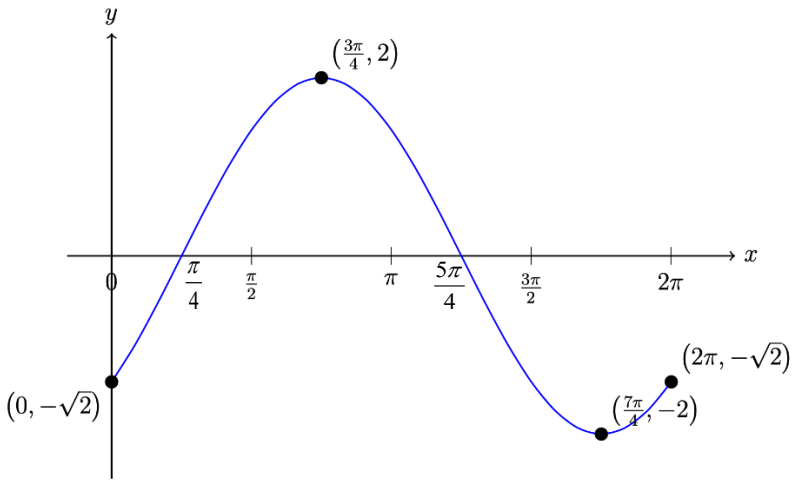
RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

H2 Math Year 6 Mock Paper 2: Solutions with Comments

No.	Suggested Solution	Remarks for Student
1 (a)	Using the identity, we have $\mathrm{f}(x)=\cos\left(x-\frac{3\pi}{4}\right)-\cos\left(x+\frac{\pi}{4}\right)$ $=-2\sin\left(\frac{2x-\frac{2\pi}{4}}{2}\right)\sin\left(-\frac{\pi}{2}\right)$ $=2\sin\left(x-\frac{\pi}{4}\right)$  $y = 2\sin\left(x - \frac{\pi}{4}\right)$ The range of f is $[-2, 2]$.	Even though the question did not specify labelling the coordinates of the stationary points, these are important features which you should mark out, as it affects the range of the function, as well as the domain restriction in (b).
(b) (i)	The largest $\alpha=\frac{3\pi}{4}$. This is because for every horizontal line $y=k,\ -\sqrt{2}\le k\le 2$, it cuts the graph of g at one and only one point. This is no longer true if $\alpha>\frac{3\pi}{4}$.	
(ii)	If $y=2\sin\left(x-\frac{\pi}{4}\right)$ then $x=\frac{\pi}{4}+\sin^{-1}\left(\frac{y}{2}\right)$. Hence $\mathrm{g}^{-1}:x\mapsto\frac{\pi}{4}+\sin^{-1}\left(\frac{x}{2}\right), x\in\mathbb{R}, -\sqrt{2}\le x\le 2.$	Similar form means that the domain of the inverse needs to be indicated like how the domain of g was indicated in the question.

No.	Suggested Solution	Remarks for Student
2 (a)	Method 1 Area of triangle $ABO = \frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ $\mathbf{a}+2\mathbf{b}+\lambda\mathbf{c}=\mathbf{0}$ $\mathbf{c} = -\frac{\mathbf{a}+2\mathbf{b}}{\lambda}$ $\overrightarrow{CA} = \mathbf{a}-\mathbf{c} = \mathbf{a}+\frac{\mathbf{a}+2\mathbf{b}}{\lambda} = \frac{(\lambda+1)\mathbf{a}+2\mathbf{b}}{\lambda}$ and $\overrightarrow{CB} = \mathbf{b}-\mathbf{c} = \mathbf{b}+\frac{\mathbf{a}+2\mathbf{b}}{\lambda} = \frac{\mathbf{a}+(\lambda+2)\mathbf{b}}{\lambda}$ Area of ΔABC $= \frac{1}{2}\left\lvert\frac{(\lambda+1)\mathbf{a}+2\mathbf{b}}{\lambda}\times\frac{\mathbf{a}+(\lambda+2)\mathbf{b}}{\lambda}\right\rvert$ $= \frac{1}{2}\left\lvert\frac{(\lambda+1)\mathbf{a}\times\mathbf{a}+2\mathbf{b}\times\mathbf{a}+(\lambda+1)\mathbf{a}\times(\lambda+2)\mathbf{b}+2\mathbf{b}\times(\lambda+2)\mathbf{b}}{\lambda^2}\right\rvert$ $= \frac{1}{2}\left\lvert\frac{\mathbf{0}-2(\mathbf{a}\times\mathbf{b})+(\lambda^2+3\lambda+2)(\mathbf{a}\times\mathbf{b})+\mathbf{0}}{\lambda^2}\right\rvert$ $= \frac{1}{2}\left\lvert\frac{(\lambda^2+3\lambda)(\mathbf{a}\times\mathbf{b})}{\lambda^2}\right\rvert$ $= \frac{1}{2}\left\lvert\frac{\lambda+3}{\lambda}\right\rvert\lvert\mathbf{a}\times\mathbf{b}\rvert$ $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ABO} = \frac{\frac{1}{2}\left\lvert\frac{\lambda+3}{\lambda}\right\rvert\lvert\mathbf{a}\times\mathbf{b}\rvert}{\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert} = \left\lvert\frac{\lambda+3}{\lambda}\right\rvert$ (shown) Method 2 $\mathbf{a}+2\mathbf{b}+\lambda\mathbf{c}=\mathbf{0}$ $\mathbf{a} = -2\mathbf{b}-\lambda\mathbf{c}$	We use properties of the cross product $\mathbf{a}\times\mathbf{a}=\mathbf{0}$, $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$ which follows from the definition $\mathbf{a}\times\mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta\hat{\mathbf{n}}$

	Area of triangle $\Delta ABO = \frac{1}{2}\left\|\mathbf{a}\times\mathbf{b}\right\|$ $= \frac{1}{2}\left\|(-2\mathbf{b}-\lambda\mathbf{c})\times\mathbf{b}\right\|$ $= \frac{1}{2}\left\|\mathbf{0}-\lambda\mathbf{c}\times\mathbf{b}\right\|$ $= \frac{1}{2}\left\|\lambda(\mathbf{b}\times\mathbf{c})\right\|$ $= \frac{1}{2}\left\|\lambda\right\|\left\|\mathbf{b}\times\mathbf{c}\right\|$ Area of $\Delta ABC = \frac{1}{2}\left\|\overrightarrow{BA}\times\overrightarrow{BC}\right\|$ $= \frac{1}{2}\left\|(\mathbf{a}-\mathbf{b})\times(\mathbf{c}-\mathbf{b})\right\|$ $= \frac{1}{2}\left\|(-2\mathbf{b}-\lambda\mathbf{c}-\mathbf{b})\times(\mathbf{c}-\mathbf{b})\right\|$ $= \frac{1}{2}\left\|(-3\mathbf{b}-\lambda\mathbf{c})\times(\mathbf{c}-\mathbf{b})\right\|$ $= \frac{1}{2}\left\|-3\mathbf{b}\times\mathbf{c}-\lambda\mathbf{c}\times\mathbf{c}-3\mathbf{b}\times(-\mathbf{b})-\lambda\mathbf{c}\times(-\mathbf{b})\right\|$ $= \frac{1}{2}\left\|-3\mathbf{b}\times\mathbf{c}-\mathbf{0}-\mathbf{0}-\lambda\mathbf{b}\times\mathbf{c}\right\|$ $= \frac{1}{2}\left\|-3-\lambda\right\|\left\|\mathbf{b}\times\mathbf{c}\right\|$ $= \frac{1}{2}\left\|\lambda+3\right\|\left\|\mathbf{b}\times\mathbf{c}\right\|$ $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ABO} = \frac{\frac{1}{2}\left\|\lambda+3\right\|\left\|\mathbf{b}\times\mathbf{c}\right\|}{\frac{1}{2}\left\|\lambda\right\|\left\|\mathbf{b}\times\mathbf{c}\right\|} = \left\|\frac{\lambda+3}{\lambda}\right\|$ (shown)	
(b)	For A, B and C to be collinear, the area of ΔABC must be 0, so $\lambda = -3$.	Try to use the result in (a) since this is only 1 mark.
(c)	Since D is on AB, by Ratio Theorem, $\overrightarrow{OD} = (1-k)\mathbf{a}+k\mathbf{b}$ when $AD:DB = k:(1-k)$. Since D lies on the line OC, $\overrightarrow{OD} = m\overrightarrow{OC} = -\frac{m}{\lambda}(\mathbf{a}+2\mathbf{b})$. We observe that the coefficient of **b** is twice the coefficient of **a**.	

	$k = 2(1-k)$ $k = \frac{2}{3}$ So $AD : DB = \frac{2}{3} : \left(1 - \frac{2}{3}\right) = 2:1$ $\therefore \overrightarrow{OD} = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$	
(d)	$\angle OAC = 90°$ $\overrightarrow{OA} \cdot \overrightarrow{CA} = 0$ $\mathbf{a} \cdot \frac{(\lambda+1)\mathbf{a} + 2\mathbf{b}}{\lambda} = 0$ $(\lambda+1)\mathbf{a}\cdot\mathbf{a} + \mathbf{a}\cdot(2\mathbf{b}) = 0$ $(\lambda+1)\lvert\mathbf{a}\rvert^2 + 2\mathbf{a}\cdot\mathbf{b} = 0$ ...(1) $\angle OBC = 90°$ $\overrightarrow{OB} \cdot \overrightarrow{CB} = 0$ $\mathbf{b} \cdot \frac{\mathbf{a} + (\lambda+2)\mathbf{b}}{\lambda} = 0$ $\mathbf{b}\cdot\mathbf{a} + \mathbf{b}\cdot\left[(\lambda+2)\mathbf{b}\right] = 0$ $\mathbf{a}\cdot\mathbf{b} + (\lambda+2)\lvert\mathbf{b}\rvert^2 = 0$ ...(2) To eliminate $\mathbf{a}\cdot\mathbf{b}$, $(1)-(2)\times 2$, $(\lambda+1)\lvert\mathbf{a}\rvert^2 - 2(\lambda+2)\lvert\mathbf{b}\rvert^2 = 0$ $\lambda\lvert\mathbf{a}\rvert^2 + \lvert\mathbf{a}\rvert^2 - 2\lambda\lvert\mathbf{b}\rvert^2 - 4\lvert\mathbf{b}\rvert^2 = 0$ $\left(\lvert\mathbf{a}\rvert^2 - 2\lvert\mathbf{b}\rvert^2\right)\lambda = 4\lvert\mathbf{b}\rvert^2 - \lvert\mathbf{a}\rvert^2$ $\lambda = \frac{4\lvert\mathbf{b}\rvert^2 - \lvert\mathbf{a}\rvert^2}{\lvert\mathbf{a}\rvert^2 - 2\lvert\mathbf{b}\rvert^2}$	

No.	Suggested Solution	Remarks for Student
3 (a)	$S_n - S_{n-1} = 75 - \frac{3^{n+1}}{5^{n-2}} - \left[75 - \frac{3^n}{5^{n-3}}\right]$ $u_n = \frac{3^n}{5^{n-3}} - \frac{3^{n+1}}{5^{n-2}}$ $= \frac{3^{n-1}}{5^{n-1}}\left(\frac{3}{5^{-2}} - \frac{3^2}{5^{-1}}\right)$ $= \left(\frac{3}{5}\right)^{n-1}(75-45)$ $= 30\left(\frac{3}{5}\right)^{n-1}, n \geq 2.$ When $n = 1$, $u_1 = S_1 = 75 - \frac{9}{5^{-1}} = 30$ which is of the same form as $30\left(\frac{3}{5}\right)^{n-1}$. Hence $u_n = 30\left(\frac{3}{5}\right)^{n-1}$ for all positive integers n. Since $\frac{u_n}{u_{n-1}} = \frac{30\left(\frac{3}{5}\right)^{n-1}}{30\left(\frac{3}{5}\right)^{n-2}} = \frac{3}{5}$ for all $n \geq 2$, the sequence is a GP with common ratio $r = \frac{3}{5}$ and first term $a = 30$. Since $\frac{3}{5} < 1$, the geometric sequence converges and $S_\infty = \frac{30}{1 - \frac{3}{5}} = 75$ **Alternative for finding sum to infinity:** $S_n = 75 - \frac{3^{n+1}}{5^{n-2}} = 75 - 27\left(\frac{3}{5}\right)^{n-2}$ As $n \to \infty$, $\left(\frac{3}{5}\right)^{n-2} \to 0$. Hence, $S_\infty = 75$	To show that the sequence is a GP, we need to show that the ratio between **<u>any two</u>** consecutive terms is a constant. The result $u_n = S_n - S_{n-1}$ holds only for $n > 1$ since S_0 does not exist. Checking a finite number of consecutive terms will not result in full credit.
(b) (i)	$v_{3k+4} = v_{3k+3} - 2 = (v_{3k+2} - 2) - 2 = (v_{3k+1} - 2) - 4 = v_{3k+1} - 6$ $\Rightarrow v_{3k+4} - v_{3k+1} = -6$ which is a constant for $k = 0, 1, 2, \cdots$. Hence the sequence v_1, v_4, v_7, ... is an arithmetic sequence.	Similarly, we need to show that the difference between **<u>any 2</u>** consecutive terms is –6.

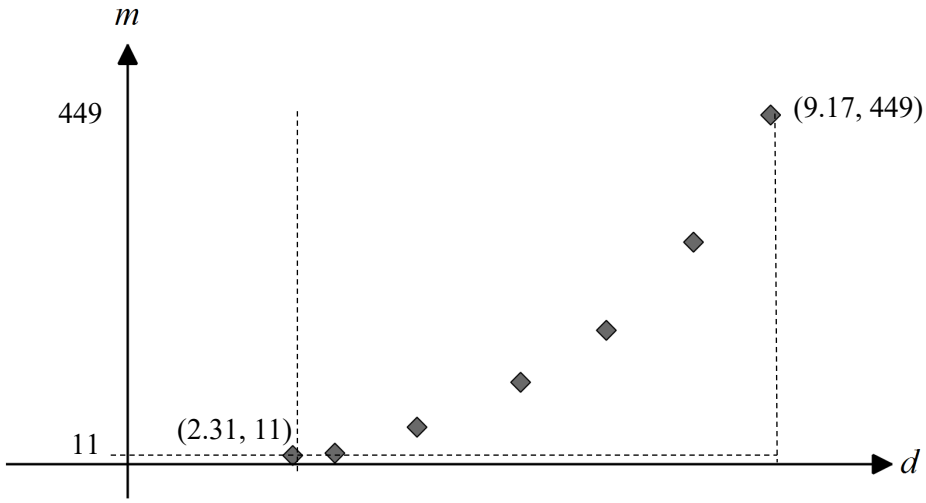
(b) **(ii)**	$\sum_{r=1}^{n} v_{3r+1} = v_4 + v_7 + \cdots + v_{3n+1}$ is an arithmetic sequence with 1st term, -3, and common difference, -6. $\sum_{r=1}^{n} v_{3r+1} < -2025$ $\frac{n}{2}[2(-3) + (n-1)(-6)] < -2025$ $n(-6n) < -4050$ $n^2 > 675$ $n > 25.98$ Hence the smallest integer n is 26.	

No.	Suggested Solution	Remarks for Student
4 (a) (i)	$y = \tan\left(e^{2x} - 1\right)$ $\frac{dy}{dx} = 2e^{2x}\sec^2\left(e^{2x}-1\right) = 2e^{2x}\left[1+\tan^2\left(e^{2x}-1\right)\right] = 2e^{2x}\left(1+y^2\right)$ where $k = 2$. (shown)	
(a) (ii)	Differentiating with respect to x, $\frac{d^2y}{dx^2} = 2e^{2x}\left(2y\frac{dy}{dx}\right) + 4e^{2x}\left(1+y^2\right)$ $\frac{d^2y}{dx^2} = 4e^{2x}\left(1+y\frac{dy}{dx}+y^2\right)$ Differentiating with respect to x, $\frac{d^3y}{dx^3} = 8e^{2x}\left(1+y\frac{dy}{dx}+y^2\right) + 4e^{2x}\left(y\frac{d^2y}{dx^2} + \left[\frac{dy}{dx}\right]^2 + 2y\frac{dy}{dx}\right)$ When $x = 0$, $y = 0$, $\frac{dy}{dx} = 2$, $\frac{d^2y}{dx^2} = 4$, $\frac{d^3y}{dx^3} = 24$	This really is an exercise in accuracy, especially when applying the chain rule and product rule.
(b)	$\tan\left(e^{2x}-1\right) \approx 0 + 2x + \frac{4}{2}x^2 + \frac{24}{6}x^3 + \ldots$ $= 2x + 2x^2 + 4x^3 + \ldots$	
(c)	$e^{ax}\ln(1+nx) = \left(1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \ldots\right)\left(nx - \frac{(nx)^2}{2} + \frac{(nx)^3}{3} + \ldots\right)$ $= nx + \left[(ax)(nx) - \frac{(nx)^2}{2}\right] + \left[(ax)\left(-\frac{(nx)^2}{2}\right) + \frac{(nx)^3}{3} + (nx)\left(\frac{(ax)^2}{2!}\right)\right] + \ldots$ $= nx + \left(an - \frac{n^2}{2}\right)x^2 + \left(-\frac{an^2}{2} + \frac{n^3}{3} + \frac{a^2n}{2}\right)x^3 + \cdots$ $\tan\left(e^{2x}-1\right) = 2x + 2x^2 + 4x^3 + \ldots$ Comparing the coefficients of x: $n = 2$ x^2: $2a - \frac{(2)^2}{2} = 2 \Rightarrow a = 2$ Coefficient of $x^3 = -\frac{2(2^2)}{2} + \frac{2^3}{3} + \frac{2^2(2)}{2!} = \frac{8}{3}$ Hence, the third non-zero term is $\frac{8}{3}x^3$.	There are 3 contributions to the coefficient of x^3. Collecting them in a systematic manner is key.

Section B: Probability and Statistics [60 marks]

No.	Suggested Solution	Remarks for Student
5 **(a)**	Number of ways of arranging the 6 couples = 6! Number of ways of arranging the husband and wife within each of the 6 couples = 2^6 Required number of ways = $= 6! \times 2^6 = 46080$.	Remember to consider the arrangement within each of the 6 groups
(b)	**Case 1** : wives sit together at one end and husbands sit at the other end [6 Ws] [6 Hs] Number of ways to arrange the 6 wives within the unit $= 6!$ Number of ways to arrange the 2 units of 6 people each = 2! Number of choices for the husband whose wife is seated next to the unit of husbands = 5 Number of ways to sit the remaining 5 men = 5! $\therefore$ Number of ways $= 6! \times 5 \times 5! \times 2 = 864000$ **Case 2** : wives sit together, but not at either end of the row A unit then consists of 2 husbands and 6 wives, with the husbands at the extreme ends of the unit. Number of ways to arrange the 6 wives within the unit $= 6!$ Number of ways to sit 2 husbands beside the wives $= (1 \times 5) + (4 \times 4) = 21$ [1 | $W_a W_b W_c W_d W_e W_f$ | 5] (H_f) [4 | $W_a W_b W_c W_d W_e W_f$ | 4] (not H_f) Number of ways to arrange this unit and the remaining 4 men $= (4+1)!$ Number of ways $= 6! \times 21 \times 5! = 1814400$ Required number of ways $= 864000 + 1814400 = 2678400$	For this question, due to the restriction that no woman is to sit next to her husband, it may be useful to approach this question by thinking about the possible identity of the man (if any) seated next to the woman on the far end of the unit of 6 women. If the man sitting to the left of the wives is the husband to the right-most wife, all the remaining 5 men can be chosen to sit on the right of the wives. In all other cases, only 4 men can sit on the right of the wives (excluding the husband of the right-most wife).

	Alternative method (Complement): Consider the 6 wives as a unit. Number of ways with wives sitting together, and left-most wife sitting with her husband $= 6!\times(5+1)!= 518400$ ------- (A) Number of ways with wives sitting together, and right-most wife sitting with her husband $= 6!\times(5+1)!= 518400$ -------- (B) Number of ways with wives together, and <u>both</u> left- and right-most wife sitting with their husbands $= 6!\times(4+1)!= 86400$ ------- (C) Number of ways with wives sitting together, and <u>either</u> left- or right-most wife sitting with her husband $= A + B - C = 518400 + 518400 - 86400 = 950400$ Number of ways with wives sitting together $= 6!\times(6+1)!= 3628800$ Required number of ways $= 3628800 - 950400 = 2678400$	(C) is included in both (A) and (B), so we have to remove a copy of it.
(c)	**Case 1** : Exactly 1 wife in the committee Number of ways $= {}^6C_1 \times {}^5C_2 = 60$ **Case 2** : Exactly 2 wives in the committee Number of ways $= {}^6C_2 \times {}^4C_1 = 60$ **Case 3** : 3 wives in the committee Number of ways $= {}^6C_3 = 20$ Required number of ways = $60 + 60 + 20 = 140$ **Alternative method (Complement):** Number of ways without restriction $= {}^{12}C_3 = 220$ Number of ways such that there are no wives $= {}^6C_3 = 20$ Number of ways such that there is exactly a couple included $= {}^6C_1 \times {}^{10}C_1 = 60$ Required number of ways = $220 - 20 - 60 = 140$	Note that the case with no wives and the case with a couple are mutually exclusive

No.	Suggested Solution	Remarks for Student
6 (a) (i)		Remember to label the axes and indicate the range of values of m and d.
(a) (ii)	Since m increases at an increasing rate as d increases, the relationship between m and d should not be modelled by a linear equation of the form $y = ax + b$.	As question asked with reference to the scatter diagram, the r value should not be considered.
(b)	For the equation: $m = ed^2 + f, r = 0.98889$ (5 s.f.) For the equation : $m = gd^3 + h, r = 0.99951$ (5 s.f.) Since the magnitude of r for the equation: $m = gd^3 + h$ is closer to 1 than the other formula, the model using the equation: $m = gd^3 + h$ is a better model. Therefore, $m = 0.572d^3 + 3.74$ (3 s.f.) Least squares estimates of g is 0.572 and h is 3.74.	The 2 models exhibit the same trend, hence when comparing them, the r-value is required.
(c) (i)	The mass in kg of a giant pumpkin with over the top length 6 m = 127.22 = 127 (3 s.f.) The mass in kg of a giant pumpkin with over the top length 12 m = 991.55 = 992 (3 s.f.)	
(c) (ii)	Since 6 lies within the range of data values of d which is (2.31, 9.17) but 12 does not, the estimate in **(A)** is more reliable.	

No.	Suggested Solution	Remarks for Student
7 **(a)**	We assume that (i) The probability of a sweet being yellow remains constant at 0.1 for all sweets. (ii) The event that a sweet is yellow is independent of the event that another sweet is yellow. Let X be the **number** of sweets out of 6 in the packet that are yellow. $X \sim \text{B}(6, 0.1)$ $\text{P}(X \le 1) = 0.885735$	0.885735 is exact
(b)	Let Y be the **number** of packets out of 90 that contain no more than one yellow sweet. $Y \sim \text{B}(90, 0.885735)$ $\text{P}(Y \ge 80) = 1 - \text{P}(Y \le 79) = 0.546$ (3 s.f.)	The accuracy of the required probability will be affected if the value of $\text{P}(X \le 1)$ used is less than 5 sf i.e. you need to either work with 0.885735 (exact) or 0.88574 (5sf)
(c)	Let R be the **number** of sweets out of 6 sweets in the packet that are red. $R \sim \text{B}(6, p)$ Since mode of R is 2, $\text{P}(R = 2) > \text{P}(R = 3)$ and $\text{P}(R = 2) > \text{P}(R = 1)$ $\text{P}(R = 2) > \text{P}(R = 3)$ $\binom{6}{2} p^2 (1-p)^4 > \binom{6}{3} p^3 (1-p)^3$ $15p^2(1-p)^4 > 20p^3(1-p)^3$ $(1-p) > \frac{4}{3}p$ since $0 < p < 1 \Rightarrow 0 < (1-p) < 1$ $\left(\frac{4}{3}+1\right)p < 1$ $p < \frac{3}{7}$	Note that the formula can be found in MF27. As the question requires the **exact** range of values, all working must be shown. Do not use the table of values from GC as working.

$$\mathrm{P}(R=2) > \mathrm{P}(R=1)$$

$$\binom{6}{2} p^2 (1-p)^4 > \binom{6}{1} p(1-p)^5$$

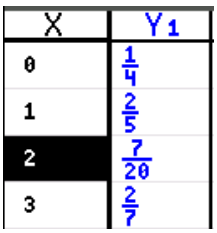
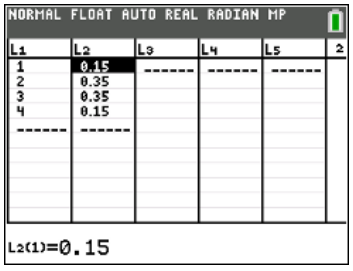
$$15p^2(1-p)^4 > 6p(1-p)^5$$

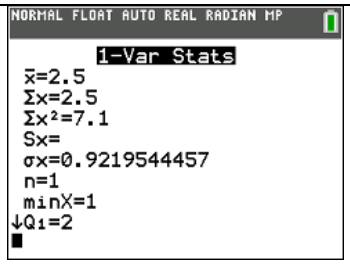
$$\frac{5}{2}p > (1-p) \quad \text{since } p > 0 \text{ and } (1-p) > 0$$

$$\left(\frac{5}{2}+1\right)p > 1$$

$$p > \frac{2}{7}$$

Therefore, $\frac{2}{7} < p < \frac{3}{7}$

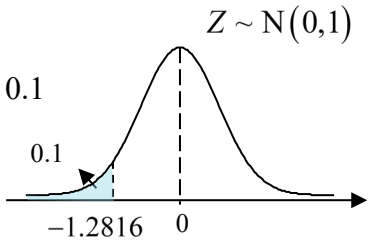
No.	Suggested Solution	Remarks for Student
8 (a)	For S to be 3, either 1 blue, 1 red and 1 yellow are chosen or 3 blues are chosen $\mathrm{P}(1R,1B,1Y)=\frac{3}{y+4}\times\frac{1}{y+3}\times\frac{y}{y+2}\times 3!$ $=\frac{18y}{(y+4)(y+3)(y+2)}$ $\mathrm{P}(3B)=\frac{3}{y+4}\times\frac{2}{y+3}\times\frac{1}{y+2}=\frac{6}{(y+4)(y+3)(y+2)}$ $\mathrm{P}(S=3)=\frac{18y}{(y+4)(y+3)(y+2)}+\frac{6}{(y+4)(y+3)(y+2)}$ $=\frac{18y+6}{(y+4)(y+3)(y+2)}$ $=\frac{6(3y+1)}{(y+4)(y+3)(y+2)}$ (Shown)	
(b)	$\mathrm{P}(S=3)=\frac{6(3y+1)}{(y+4)(y+3)(y+2)}=\frac{7}{20}$ Using GC, since y is a positive integer, $y=2$. $\mathrm{P}(S=1)=\mathrm{P}(1B,2Y)=\frac{3}{6}\times\frac{2}{5}\times\frac{1}{4}\times\frac{3!}{2!}=\frac{18}{120}=\frac{3}{20}$ $\mathrm{P}(S=2)=\mathrm{P}(2B,1Y)+\mathrm{P}(1R,2Y)$ $=\frac{3}{6}\times\frac{2}{5}\times\frac{2}{4}\times\frac{3!}{2!}+\frac{1}{6}\times\frac{2}{5}\times\frac{1}{4}\times\frac{3!}{2!}=\frac{7}{20}$ $\mathrm{P}(S=4)=\mathrm{P}(2B,1R)=\frac{3}{6}\times\frac{2}{5}\times\frac{1}{4}\times\frac{3!}{2!}=\frac{18}{120}=\frac{3}{20}$ s | 1 | 2 | 3 | 4 $\mathrm{P}(S=s)$ | $\frac{3}{20}$ | $\frac{7}{20}$ | $\frac{7}{20}$ | $\frac{3}{20}$	Key $Y_1=\frac{6(3x+1)}{(x+4)(x+3)(x+2)}$ 
(c)	From GC, $\mathrm{E}(S)=2.5$ (this is also observed by symmetry) $\mathrm{Var}(S)=7.1-2.5^2=0.85$ (exact)	Using 1-Var Stat 

		Note that using 0.921954457^2 gives 0.8499999999, which you can suspect is 0.85, given also how $\mathrm{E}\left(S^2\right)=7.1$ is exact

No.	Suggested Solution	Remarks for Student
9 (a)	Let X be the radius of a metal bolt in cm. $X \sim \mathrm{N}\left(0.8,\ 0.01^2\right)$ $\mathrm{P}\left(0.79 < X < 0.82\right) = 0.81859$ (5 s.f.) Required percentage is 81.9%. (3 s.f.)	Question asked for %
(b)	Let Y be the inside radius of a washer in cm. $Y \sim \mathrm{N}\left(0.81,\ 0.012^2\right)$ Let C be the inside circumference of a washer in cm where $C = 2\pi Y$. $\mathrm{E}(C) = 2\pi \mathrm{E}(Y) = 2\pi(0.81) \approx 5.0894 = 5.09$ (3 s.f) $\mathrm{Var}(C) = (2\pi)^2\ \mathrm{Var}(Y) = (2\pi)^2 (0.012)^2 \approx 0.0056849 = 0.00568$ (3 s.f) $C \sim \mathrm{N}\left(5.09,\ 0.00568\right)$ To find c such that $\mathrm{P}\left(C > c\right) = 0.05$ From GC, $\mathrm{P}\left(C > 5.21\right) = 0.05$ $\therefore c = 5.21$ (3 s.f.)	Use the 5 s.f answers for the parameters of C for intermediate working!
(c)	$\mathrm{E}(Y - X) = \mathrm{E}(Y) - \mathrm{E}(X) = 0.81 - 0.8 = 0.01$ $\mathrm{Var}(Y - X) = \mathrm{Var}(Y) + \mathrm{Var}(X) = 0.012^2 + 0.01^2 = 0.000244$ $Y - X \sim \mathrm{N}\left(0.01,\ 0.000244\right)$ $\mathrm{P}\left(0 < Y - X < 0.04\right) = 0.71158$ (5 s.f.) $= 0.712$ (3 s.f.)	
(d)	Let T be the outside radius of a washer in cm. $T \sim \mathrm{N}\left(\mu,\ \sigma^2\right)$ $\mathrm{P}\left(T > 1.25\right) = 0.15$ $\mathrm{P}\left(Z > \dfrac{1.25 - \mu}{\sigma}\right) = 0.15$ From GC, $\mathrm{P}\left(Z > 1.0364\right) = 0.15$ $\dfrac{1.25 - \mu}{\sigma} = 1.0364$ $\mu + 1.0364\sigma = 1.25 \quad ---(1)$	

	$\mathrm{P}(T < 1.15) = 0.25$ $\mathrm{P}\left(Z < \frac{1.15-\mu}{\sigma}\right) = 0.25$ From GC, $\mathrm{P}(Z < -0.67449) = 0.25$ $\frac{1.15-\mu}{\sigma} = -0.67449$ $\mu - 0.67449\sigma = 1.15 \quad ---(2)$ Using GC, $\mu = 1.19, \quad \sigma = 0.0584$ (3 s.f.)	

No.	Suggested Solution	Remarks for Student
10 (a)	Since the Production Manager is looking for a change in the average time which can be more or less than the current one, he should use a 2-tail test.	
(b)	Since the sample size of 50 is large, by Central Limit Theorem, the distribution of sample mean time taken to manufacture a control panel will follow a normal distribution approximately.	It is incorrect to say that T follows a normal distribution approximately due to Central Limit Theorem
(c) (i)	Unbiased estimate of the population mean $\bar{x}$ $= \frac{835.7}{50} = 16.714$ Unbiased estimate of the population variance, $s^2 = \frac{1}{50-1}\left(14067.17 - \frac{(835.7)^2}{50}\right) \approx 2.0261 = 2.03$ (3 s.f.)	
(ii)	Let T be the time taken to manufacture an electronic control panel using the alternative process. Let μ denote the **population mean time** taken to manufacture an electronic control panel. Null Hypothesis $H_0 : \mu = 17$ Alternative Hypothesis $H_1 : \mu \neq 17$ Perform a two-tail test at 10% level of significance Under H_0, since n = 50 is large, by Central Limit Theorem, $\bar{T} \sim \mathrm{N}\left(17, \frac{2.0261}{50}\right)$ approximately. Using a z-test, p-value = $2\mathrm{P}\left(\bar{T} < 16.714\right) \approx 0.155$ Since p-value = 0.155387 > 0.10, we do not reject H_0 and conclude that there is insufficient evidence at 10% level of significance that the **population** mean time taken in hours for the manufacture of a control panel has changed.	

(d)	Let the population variance of the time taken to manufacture a control panel be σ^2. $H_0 : \mu = 17$ vs $H_1 : \mu < 17$ Perform a one-tail test at 10% level of significance Under H_0, since $n = 40$ is large, by Central Limit Theorem, $\overline{T} \sim \mathrm{N}\left(17, \frac{\sigma^2}{40}\right)$ approximately. Since there is no reason to reject H_0, p-value = $\mathrm{P}\left(\overline{X} \leq 16.7\right) > 0.10$ $\mathrm{P}\left(Z \leq \frac{16.7-17}{\frac{\sigma}{\sqrt{40}}}\right) > 0.1$ From GC, $\mathrm{P}\left(Z \leq -1.2816\right) = 0.1$ From the diagram,  $-\frac{0.3\sqrt{40}}{\sigma} > -1.2816$ $\sigma > 1.4805$ $\sigma^2 > 2.19$ (3 s.f.) If the population variance of the times is more than 2.19 hours^2, the Finance Manager will conclude that there is no reason to reject the null hypothesis.	Note that the question asks for population variance, not its standard deviation.